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End Semester Examination 2024

Name of the Course: B. Tech (Civil Engineering)

Name of the Paper: Building Materials and Construction Equipments

Time: 3 Hours

Semester: III

Paper Code: TCE 304

Maximum Marks: 100

Note:

All questions are compulsory

Answer any two sub questions among a, b & c in each main questions

Total marks in each question are twenty. Each question carries 10 marks

- Q1** (20 marks)
- (a) Write short note on Air entrained concrete, Fiber reinforced concrete.
 - (b) Explain different tests on concrete.
 - (c) What is the effect of water cement ratio on the strength of concrete? **CO1**
- Q2** (20 marks)
- (a) What do you understand by synthetic aggregates? Explain its manufacturing process.
 - (b) What do you mean by binders? Explain sulphur extended asphalt, Bio derived asphalt binders and high modify asphalt binders. **CO2**
 - (c) What do you understand by energetically modified cement?
- Q3** (20 marks)
- (a) What do you mean by Magnetosrictive materials.
 - (b) Write about Electrorheological Fluids. **CO3**
 - (c) What is the purpose of cellulose, cork, Non-VOC paints, Fiber cement.
- Q4** (20 marks)
- (a) With a neat sketch explain the components of a scraper.
 - (b) Explain about the types of material handling equipments and explain with a neat sketch about any four material handling equipments? **CO4**
 - (c) Enumerate in detail about different types trucks.
- Q5** (20 marks)
- (a) Write in detail about the various methods for dewatering
 - (b) Explain in detail about the equipments needed for dewatering and pumping. **CO5**
 - (c) Explain in detail about different types of cranes.